

Coupling-gated cortical coherence: what a field theory of conscious unity must assume, and what it predicts

Sebastian Bobadilla-Suarez
Independent Researcher
`sebastian.bobadilla.s@gmail.com`
ORCID: 0000-0002-5951-0772

6 September 2026

Abstract

A field account of cortical coordination holds that spatially distributed neural processes are coupled in part through the endogenous extracellular electromagnetic field. We ask what such an account must assume to reach a globally consistent, self-referential state, and what it predicts before it gets there. Its physical core is a noisy mean-field phase model whose coupling drifts through a synchronization threshold. If recovery of coherence after sleep is gated by that crossing, it is not free to take an arbitrary time course. For as long as the phase distribution tracks the stationary branch, it must show a square-root onset and a von Mises concentration–coherence relation with no fitted shape parameter. The tracking condition is part of the prediction, and the simulations measure its cost: ramp speed sets the apparent onset exponent, so an exponent fitted to a driven recovery trace reports the drive as much as the mechanism. A coupling plasticity that descends a squared-drift objective removes most of the objective available to it and keeps the system coherent, while driving within-cluster coupling far below between-cluster coupling—away from the structure it

is asked to learn, and further from it than from almost every relabelling of the nodes; two controls sharing its step size, the second of them also its cumulative kernel deformation, do neither, and against partitions of the same nodes drawn blind to frequency the loss does not appear at all. Descending that functional does not produce representational learning: for this pairing of objective and template the optimum is opposed to the target by a conservation law rather than by accident. A Lean 4 formalization composes finite capacity, Landauer’s bound, the noisy Kuramoto transition, sheaf gluing and Banach’s theorem into one conditional theorem, isolating the eight hypotheses it consumes and the two structures it takes alongside them, whose class fields carry physical content of their own; one witness assembled from distinct toy components satisfies all eight at once. We defend as a philosophical commitment, not a theorem, the identification of the glued section with the unity of a conscious episode and of its reflexive fixed point with a minimal self. Testing the account requires independent calibration of effective cortical coupling and an awakening experiment separating slow coupling change from alternative recovery mechanisms.

1 Introduction

Neural activity changes across sleep, awakening and anesthesia, but a change in synchrony does not by itself identify the physical interaction responsible for it. A field account makes a stronger proposal: spatially distributed neural processes are coordinated partly through the endogenous extracellular electromagnetic field, and changes in the medium can change that coordination (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Whether that mechanism is sufficient to support the collective state such accounts require remains an empirical question.

11 Such an account has three requirements, and they order what follows.
12 Coupling must be strong enough to sustain coherent collective activity; the
13 local descriptions that activity carries must be compatible enough to glue
14 into a single one; and that glued state must support an accurate model of
15 itself. The first requirement is physical and testable with current methods,
16 and most of this article is about testing it. The second and third are where
17 the mathematics is, and where the assumptions accumulate.

18 The first requirement admits a definite claim. If cortical coupling is slowly
19 gated—if recovery of coherence after sleep is driven by a coupling parameter
20 drifting through a synchronization threshold—then the recovery is not free
21 to take an arbitrary time course. For a recovery slow enough that the phase
22 distribution tracks the stationary branch, the theory fixes a square-root onset
23 and a one-parameter relation between phase concentration and coherence,
24 with no fitted shape parameter. That is a constraint an experiment can
25 meet or violate. How slow is slow enough is itself part of the claim, and
26 our simulations bound it from the other side, by showing which parts of the
27 constraint survive finite populations and finite ramp speeds and which do
28 not.

29 The contribution has four parts. The first is the coupling-gated recovery
30 hypothesis itself: a stationary coherent branch with square-root onset, a von
31 Mises concentration–coherence relation, and an experimental protocol with
32 failure conditions declared in advance (Sections 2 and 3). The second is a
33 constraint on how such data may be analyzed. The apparent onset expo-

34 nent measured from a finite ramp is a function of ramp speed across the
 35 sweep of Section 4, so an exponent fitted to a recovery trace measures the
 36 ramp at least as much as the mechanism—and that holds for any analysis
 37 reading a critical exponent off a driven transition, not only for ours. The
 38 third is a negative result about plasticity: a coupling update descending a
 39 squared-drift objective removes most of the objective available to it and pre-
 40 serves coherence, while moving *away* from the environmental structure it
 41 is expected to learn—below almost every node relabelling, below all but a
 42 twentieth of the partitions drawn blind to that structure, and beyond what
 43 either of two controls does: a random update of the same step size, and that
 44 same gradient step delivered to relabelled nodes, which reproduces its cu-
 45 mulative kernel deformation. Descending a squared-drift functional does not
 46 produce representational learning, and the direction of that failure follows
 47 from a conservation law rather than from the particular run. The fourth
 48 is a machine-checked composition (Section 5) in which finite information
 49 capacity, Landauer’s bound, a predictive dissipation bound, the noisy Ku-
 50 ramoto transition, sheaf gluing and Banach’s fixed-point theorem yield one
 51 conditional theorem, with the eight hypotheses that composition consumes
 52 isolated and named; one witness assembled from distinct toy components
 53 satisfies all eight at once, so their conjunction is not empty.

54 A prose table of assumptions would list the same eight items, so the fourth
 55 contribution needs a word about method. What the Lean development adds
 56 is that it makes the list non-negotiable. The characteristic failure in an

57 argument of this shape is not a false step but a silent change of subject:
 58 a theorem about a stationary density read as a claim about a trajectory,
 59 an inequality about a register read as a claim about learning, agreement of
 60 phases read as compatibility of representations. Type-checking forbids each
 61 of those substitutions, and an assumption needed to make a step go through
 62 appears as a hypothesis in the term rather than as a word in a sentence. The
 63 last of them is not merely forbidden but refuted by a constructed object: the
 64 development carries a cover at a single common phase whose local densities
 65 disagree where its patches meet, satisfying every other requirement of the
 66 gluing theorem and failing compatibility alone (Section 5.2). That object
 67 is what a prose table of assumptions cannot carry. Formalization does not
 68 make the assumptions true; it makes them impossible to lose.

69 The physical model and the formal composition belong in one article be-
 70 cause they are two halves of one commitment. The coupling-gated prediction
 71 is worth testing under this framework, rather than as one more synchroniza-
 72 tion model, only because the coherent state it produces is claimed to do
 73 further work: to support a globally consistent representation and a reflexive
 74 fixed point that we identify with phenomenal unity and a minimal self.

75 The distinction that does that work is between phases agreeing and de-
 76 scriptions fitting together, and it is concrete. Take two cortical regions whose
 77 receptive territories overlap, each carrying a probability distribution over
 78 what occupies the shared part. Their oscillators can be locked to the same
 79 phase at every instant while one region's distribution on the overlap places an

80 object left of fixation and the other places it right. The phases agree; the de-
81 scriptions do not, and no single distribution over the shared territory restricts
82 to both. A theory that identifies unity with synchrony cannot mark that dif-
83 ference, and so cannot separate a brain whose regions oscillate together from
84 one whose regions are about the same world. The sheaf condition marks
85 exactly it: local states glue to one global section when their restrictions to
86 shared regions coincide, and otherwise not at all. Coherence enters as the
87 physical condition under which compatibility can be maintained, not as a
88 substitute for it, and the self-prediction map then asks the further ques-
89 tion of whether the glued state contains an accurate model of itself. That
90 identification is a commitment, not a theorem, and Section 6 states it, ar-
91 gues for it, and says what would count against it. Its separation from the
92 physical tests is what makes both assessable: the cortical coupling hypothe-
93 sis and the recovery prediction can fail while the mathematical composition
94 stands, and the composition can stand while the identification is wrong. The
95 established ingredients (Landauer, 1961; Still et al., 2012; Sakaguchi, 1988;
96 Banach, 1922) therefore appear here for their biological composition, not for
97 any new fundamental physical law.

2 A coupling-gated model of cortical coherence

2.1 Dynamics and stationary solutions

The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units, with angles treated as dimensionless. The uniform mean-field convention is $K_{ij} = K/N$. Frequency heterogeneity and phase diffusion are separate parameters; a frequency-spread width cannot be substituted for D when applying the identical-frequency stationary theory.

For identical frequencies in their rotating frame, the assumed stationary density is

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}. \quad (2)$$

The I_n are modified Bessel functions, and ψ is the collective phase. Lean verifies normalization, the density's circular order parameter, and the self-consistency equation. At $K \leq K_c = 2D$, the only nonnegative self-consistent order parameter is zero. At $K > K_c$, exactly one positive solution coexists with zero; the positive branch is continuous at threshold and strictly

114 increasing with K .

115 The near-threshold expansion gives

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

116 This is a result about the stationary solution branch. Of the coherent branch
117 we prove existence, uniqueness, continuous emergence at threshold and strict
118 growth in K , but not that it is dynamically selected. The dynamical mean-
119 field limit—propagation of chaos—is the missing link between the finite sys-
120 tem and the stationary description (Sznitman, 1991), and no theorem in this
121 development closes it. Exchange symmetry alone does not justify factoriza-
122 tion of the joint law, particularly for an ensemble with an unpinned collective
123 phase.

124 A spatial version replaces the finite sum by an integral against a kernel
125 $K(x, y)$. We normalize the substrate measure to total mass one, so the mean-
126 field strength is a kernel average, not a quantity that grows with the arbitrary
127 measure of the substrate. Reduction of a heterogeneous spatial kernel to a
128 scalar K is an additional modelling step, and the threshold in Eq. (2) does
129 not automatically apply to every spatial or signed network.

130 **2.2 Physical identification and calibration**

131 Regular mesh refinement gives convergence of a discrete coupling-energy
132 quadrature to its continuum integral: scalar energy convergence, not conver-

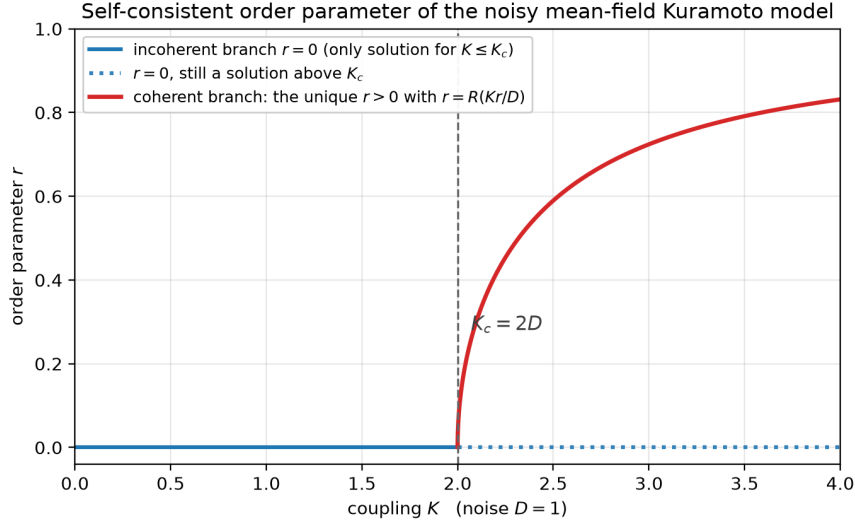


Figure 1: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the same integral expressions used in the formalization; it illustrates the static theorems and does not establish dynamical selection.

133 gence of trajectories and not a construction of a physical kernel. The model
134 posits a continuum kernel and separately identifies its cortical realization
135 with endogenous electromagnetic coupling. The formal predicate requires
136 a jointly continuous kernel, a normalized substrate, a mean-field strength
137 equal to its spatial average, and no single site providing more than half of
138 that mean-field coupling. Toy witnesses satisfy these conditions; none is a
139 model calibrated to cortex.

140 Extracellular fields can modulate spike timing (Fröhlich & McCormick,
141 2010; Anastassiou & Koch, 2015), and field-based models have been proposed

for integrating distributed neural information (Pinotsis & Miller, 2022; McFadden, 2020). These observations motivate an effective interaction without establishing its magnitude relative to phase diffusion, its spatial averaging rule, or its causal role in experience; changes in phase organization under propofol (Bastos et al., 2021) likewise do not measure a crossing of the model’s coupling threshold.

The physical-unit conversion is the sharpest open limitation of the framework. An estimate proportional to $N\Delta tf$, using population size, a spike-time shift and oscillation frequency, is dimensionless. It cannot be compared directly with an inverse-time diffusion parameter. A conversion such as

$$K_{\text{eff}} = \gamma N s E f \tag{4}$$

requires a rate calibration γ , where s is the time shift per unit field and E the field amplitude. Supplemental Material sets out why γ is not recoverable from the other factors: it states how a per-cycle phase advance accumulates into a drift rate, and the choice that would let it cancel removes the very ratio the threshold test is about. The saved field-strength calculations are conditional on this calibration and on the chosen spatial aggregation; they do not independently establish $K_{\text{eff}} > 2D$ in cortex. Determining the interaction kernel and phase diffusion from the same physical system is necessary before the threshold becomes a quantitative biological claim; Section 3.2 makes that measurement the first item of the proposed experiment.

162 3 Recovery after awakening: a conditional 163 prediction

164 Sleep-associated changes in extracellular space and clearance motivate a pos-
165 sible slow control of coupling (Xie et al., 2013). That geometry is measurable
166 and its cortical values are known: the extracellular space occupies a volume
167 fraction $\alpha \approx 0.2$ of tissue at a diffusion tortuosity $\lambda \approx 1.6$ (Syková & Nichol-
168 son, 2008), and natural sleep raises α from roughly 0.14 to 0.23 (Xie et al.,
169 2013). It reaches Eq. (4) through the field amplitude E rather than through
170 a further mechanism: below 10 kHz cortical tissue conducts approximately
171 resistively at $0.3\text{--}0.6\text{ S m}^{-1}$, an order of magnitude set by the conductiv-
172 ity of saline diluted into that volume fraction (Barbour, 2017), so at fixed
173 transmembrane current density a larger α yields a smaller extracellular field.
174 We hypothesize that recovery after awakening includes a change in effective
175 coupling $K(t)$, mediated by extracellular geometry, on a slower timescale
176 than phase dynamics. The conductivity route fixes the sign that hypothesis
177 predicts—contraction of the extracellular space on awakening raises K —so
178 the design of Section 3.2 can fail it in direction as well as in magnitude. It
179 fixes nothing further: it is an effective-medium argument at fixed current
180 density, and is silent on how a heterogeneous field aggregates into the scalar
181 K of Eq. (2). Neither the monotonic relation between geometry and cou-
182 pling nor its role in human sleep inertia follows from the cited physiology;
183 the model requires both to be measured.

184 If $K(t)$ rises through K_c and the phase distribution tracks the coherent
 185 stationary branch, Eq. (3) predicts

$$r(t) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad t > t_0, \quad \dot{K}(t_0) > 0. \quad (5)$$

186 The ideal stationary description has a subthreshold interval followed by con-
 187 tinuous onset. Saturation requires a specified subsequent coupling trajectory;
 188 monotonic coupling alone does not determine a logistic or any unique sig-
 189 moid. The onset time is fixed by $K(t_0) = K_c$ only if $K(t)$ is independently
 190 known. When coupling is inferred from the same coherence trace, this is not
 191 a parameter-free prediction of latency.

192 Critical slowing down limits the adiabatic prediction of Eq. (5) near
 193 threshold, and a finite population carries nonzero fluctuating order below
 194 it. An observable onset therefore requires a predeclared threshold above the
 195 finite-size floor, and a finite-rate trajectory need not show an infinite-slope
 196 cusp. Section 4.1 quantifies the limitation.

197 **3.1 Distributional and temporal tests**

198 The stationary model makes a second prediction: phase concentration and
 199 coherence obey $r = I_1(a)/I_0(a)$, with no fitted shape parameter. Its appli-
 200 cability depends on the stationary-density assumption and the observation
 201 model, and the relation is common to the whole von Mises family, so agree-
 202 ment with it identifies neither ephaptic coupling nor this framework against

203 alternative synchronization models. Within the same ansatz $a/r = K/D$ for
204 $r > 0$, so the limiting threshold ratio follows from the density relation instead
205 of testing it, and ratios near the finite-size floor are especially unstable.

206 The estimators must be independent, which is the constraint most easily
207 violated in practice. Estimating a by inverting the Bessel ratio from r would
208 enforce the relation being tested, and linear statistics on wrapped phases
209 estimate neither quantity.

210 **3.2 A protocol for the awakening experiment**

211 The requirements above define a design rather than a single measurement,
212 and the design is where the content of the hypothesis lives. An adequate test
213 needs all six of the following.

- 214 **1. Coupling and diffusion in the same units.** Estimate K and D
215 from the same cortical regime, in inverse-time units, separating phase
216 diffusion from quenched frequency spread and from observation noise.
217 Without this the threshold $K > 2D$ is not an evaluable statement
218 about cortex.
- 219 **2. Geometry calibrated to coupling.** A diffusion-MRI observable or
220 an astrocytic signal is an indicator of the extracellular volume fraction
221 α , not a measurement of K . Its mapping to effective coupling must be
222 established separately, with its own uncertainty, and the conductivity
223 argument of Section 3 supplies only the expected sign of that mapping.

- 224 **3. Phase estimation at adequate spatial resolution.** At least 100
225 spatial samples per estimate. Estimate r by the circular resultant
226 length and a by an unweighted regression of log-density on the co-
227 sine of phase relative to the circular mean. With fewer sites, pooling
228 over time changes the estimand to a spatiotemporal distribution and
229 must be stated as such.
- 230 **4. An independent recovery endpoint.** Behavioral recovery measured
231 separately from the phase trace, so that latency is not defined by the
232 signal whose latency is at issue.
- 233 **5. A predeclared onset criterion.** A coherence threshold fixed above
234 the finite-size floor before unblinding, chosen in the knowledge that a
235 finite ramp smears the cusp.
- 236 **6. Model comparison under matched observation.** Coupling-gated
237 recovery compared against exponential, flexible sigmoidal and other
238 slow-recovery models under identical temporal filtering and observation
239 noise. Latency alone cannot distinguish the mechanism, because many
240 processes create delays.

241 The design fails informatively in three ways: failure of the stationary
242 phase-density relation over a sufficiently broad concentration range; absence
243 of the proposed coupling–geometry association; or a recovery trace better
244 explained by an alternative after finite-rate and observation effects are ac-
245 counted for. Two apparent failures are not failures. A linear apparent onset

is evidence against the adiabatic approximation only where that approximation is independently justified, and it does not falsify the field hypothesis under arbitrary ramp speeds. The absence of a fold in the stationary equation excludes stationary coexistence of two positive branches in this model, but does not exclude finite-rate hysteresis or adjudicate theories of ignition.

4 Simulation results

The simulations characterize the finite-size, finite-rate regime that the stationary theory leaves open, and they bear on whether the prediction of Section 3 is measurable at all. Two of them, Sections 4.1 and 4.3, illustrate methodological issues that also arise in analyses of driven transitions and dissipation-driven plasticity. Each is a finite run at its sampled parameters and not a general statement, and none discharges a hypothesis of the composition in Section 5. Each run uses explicit seeds and compact saved summaries, and the numerical values below are generated from those artifacts. The supplement gives parameters, estimators, failure criteria and additional panels. Identical-frequency runs use phase diffusion rather than a quenched frequency-spread surrogate, and order plots display the $1/\sqrt{N}$ reference.

4.1 Ramp speed sets the apparent onset exponent

The dynamic-ramp study (Fig. 2) uses $N = 2000$, $D = 1$ and 32 replicas at four coupling speeds. Escape is the first of three consecutive saved samples at

266 $r \geq 0.2$ after threshold. Only 5/32 replicas escape within the fastest window.
 267 The three slower ramps have mean coupling delays 0.1947, 0.0684 and 0.0253,
 268 with a fitted delay exponent 0.443. Their apparent onset exponents run from
 269 0.898 to 0.417, against 0.443 to 0.448 returned by the same estimator on the
 270 exact stationary branch sampled on the same coupling grids. At the fastest
 271 speed the ensemble mean never reaches the declared fit window, so the value
 272 the estimator returns there, 0.964, is a local log-slope of a delayed trajectory
 273 rather than an onset exponent.

274 The exponent recovered by fitting a driven transition is a joint function
 275 of the mechanism and the drive rate. An onset exponent read off a recovery
 276 trace therefore says little about the underlying transition unless the ramp rate
 277 is independently known and the finite-size floor is declared, and that holds
 278 for any such analysis, not only for one conducted under this framework. The
 279 prediction of Section 3 is the first thing it constrains: the stationary square-
 280 root law cannot be imposed on an arbitrary recovery trace, and its absence
 281 is not by itself evidence against a coupling-gated mechanism. The specified
 282 deviation-based critical ramp speed is not bracketed by the sweep.

283 The dynamical-selection control shows escape toward a coherent branch
 284 above threshold. At $K = 2.8$, steady mean order is 0.67735 against the static
 285 value 0.68270. Refining the integration step by a factor of 4 at fixed dura-
 286 tion moves that order to 0.68168, so the bulk of the gap is Euler–Maruyama
 287 bias; the last refinement still moves it by 0.00147, so the remaining -0.00102
 288 bounds the finite-size part rather than measuring it. The fitted growth-rate

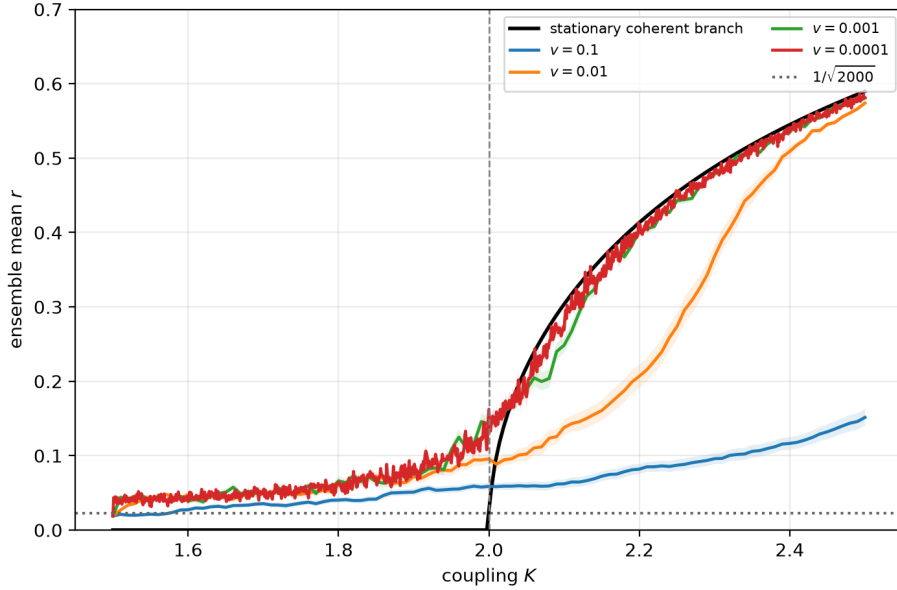


Figure 2: Finite-rate departures from the stationary branch. Ensemble-mean order is plotted against the increasing coupling for four ramp speeds v , with bands of one standard error across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape; the stationary branch is not a universal time course for recovery.

289 slope is 0.41505 against the theoretical $1/2$, over a window bounded below
 290 by the finite-size floor and above by the static branch; the same integrator
 291 bias enters that fit. The propagation-of-chaos diagnostics show decreasing
 292 subcritical dependence with increasing population size. In the supercritical
 293 unpinned ensemble, however, pair correlation remains substantial because
 294 realizations share a random collective orientation. This negative uncondi-
 295 tional result does not settle a conditional mean-field limit after removal of
 296 that orientation.

297 4.2 Spatial coupling, frustration and calibration

298 A periodic two-dimensional sheet with row-normalized exponential coupling
299 remains coherent in the sampled empirical decay-length band under the de-
300 clared total coupling and frequency spread. At decay lengths 0.1, 0.2 and
301 0.3 mm, steady order is 0.9424, 0.9427 and 0.9427. This is one finite param-
302 eter regime; fixing total coupling while changing range tests fragmentation,
303 not whether cortex supplies that total coupling.

304 A row-balanced Dale-law network shows that row balance alone does not
305 guarantee a floor-level incoherent baseline. In a separately specified weak-
306 synapse regime, a uniform modulatory term produces an operational coher-
307 ence crossing inside the bracket $K_{\text{eff}}/D \in [1.958, 2.009]$, from a coupling grid
308 refined to steps of 0.025 inside the one geometric step $[1.857, 2.110]$ that lo-
309 cates it. The refined bracket still contains the model’s own mean-field thresh-
310 old $K_c/D = 2$, so a tenfold finer grid does not separate the crossing from
311 the threshold at this size. Signed coupling amplifies finite-size fluctuations
312 in the same runs, whose incoherent baseline carries $N\langle r^2 \rangle > 1$, which is the
313 direction a resolved crossing would be displaced in; the refinement bounds
314 that displacement by the width of the bracket rather than exhibiting it. The
315 quantity K_{eff}/D is dimensionless; carrying it over to cortex needs the rate
316 calibration γ of Section 2.2, which neither this sweep nor the spatial control
317 supplies.

318 4.3 Squared-drift plasticity does not learn structure

319 The predictive dissipation bound of Section 5 bounds nonpredictive memory
320 by dissipation, and it is tempting to read a system that lowers its dissipation
321 as thereby coming to mirror the structure of its input. The bound licenses no
322 such reading: a memory carrying no information at all satisfies it with room
323 to spare, so lowering dissipation presses toward less representation rather
324 than toward better-aligned representation. This control exhibits a system in
325 which the pressure is not merely toward less but away.

326 For fixed phases, consider the squared-drift objective

$$\sigma(K) = \frac{1}{D} \int \left[\omega(x) + \int K(x, y) \sin(\theta_y - \theta_x) d\mu(y) \right]^2 d\mu(x). \quad (6)$$

327 It is a quadratic functional of an affine image of the kernel. The formalization
328 constructs the bounded drift operator A and its adjoint on the relevant L^2
329 spaces, obtaining $\nabla\sigma = (2/D)A^*(AK + \omega)$, and the specified gradient flow
330 decreases this objective at fixed phases. The objective is not generally a
331 thermodynamic entropy-production rate, which requires probability-current
332 information. Whether biological couplings follow this flow is an additional
333 modelling question.

334 The joint control evolves noisy phases with clustered frequencies and
335 nonzero grand-mean drive, while updating symmetric couplings every 50 fast
336 steps. Couplings remain nonnegative, hollow and normalized to a fixed to-
337 tal. The symmetric-edge derivative of $\frac{1}{2} \sum_i v_i^2$ is $(v_i - v_j) \sin(\theta_j - \theta_i)$, so

338 both endpoints contribute. Resource constraints and evolving phases re-
 339 move the fixed-phase monotonicity guarantee. Each seed runs four arms
 340 on one phase-noise stream: the gradient step, a symmetric random step of
 341 the same Frobenius norm, that same gradient step under a random node
 342 relabelling, and a frozen kernel. Denying an implication requires exhibiting
 343 its antecedent, so the reported quantity is how much of the objective the
 344 update actually removes: symmetric coupling cancels from the mean drift,
 345 which fixes a floor of 990 beneath the frozen arm's 1380.6063 to 1382.3448,
 346 and the gradient arm removes between 0.6773 and 0.6914 of that headroom.
 347 The random arm of matched step size removes none of it, so the descent
 348 belongs to the gradient. Matching the step size does not match the defor-
 349 mation it accumulates: successive isotropic steps cancel, leaving the random
 350 arm at a cumulative norm growth of 1.2879 to 1.3056 against the gradient
 351 arm's 12.4094 to 14.6958, so it barely enters the channel through which clip-
 352 and-rescale concentrates the fixed resource onto fewer edges. The permuted
 353 arm enters that channel. A relabelling is an isometry that leaves the multiset
 354 of step entries untouched, so the arm takes a step of the gradient's size and
 355 of its shape on edges the gradient did not select: it reaches a norm growth
 356 of 11.4185 to 16.3296, bracketing the gradient arm's, and a second-half order
 357 of 0.9184 to 0.9189, which lies inside the gradient arm's range. Its descent
 358 fraction is -0.3319 to -0.1607 : at the gradient arm's own deformation and
 359 coherence, a step that carries the gradient's entries to the wrong edges raises
 360 the objective instead of lowering it. The learning rate 0.2 is the rate at which

the objective descends furthest across a sweep from 0.001 to 0.4 evaluated on one of the three reported seeds, not a rate at which the update is known to converge.

Across three seeds, second-half order lies between 0.9167 and 0.9222: coherence survives the descent. Alignment with the cluster template is read from the ratio of mean within-cluster to mean between-cluster coupling, which is unchanged by rescaling the kernel. That ratio starts between 0.9824 and 0.9948 and ends between 0.3507 and 0.6135, while the random arm leaves it at 0.9965 to 0.9997 and the equally deformed permuted arm at 0.7290 to 1.2935, a range that contains the no-preference line and does not overlap the gradient arm's; against 2000 node relabellings the final kernel is further from the true template than from a fraction 0.9510 to 1.0000 of them (Fig. 3). The reference for that fraction is the frozen arm on the same seeds, which reaches 0.6435 to 0.9110 without any update at all, and not a null median of one half; the increment the gradient step contributes is the difference between those ranges, which at three seeds is small enough that they nearly meet. The permuted arm reaches 0.2680 to 0.8410, so what carries the gradient arm above the frozen reference is the direction of the step and not the deformation it accumulates. What the descent moves against is that partition and not block structure at large. Read against 2000 partitions of the same group sizes drawn blind to frequency, at most a fraction 0.0495 of them give a lower within-over-between ratio than the frequency partition does on any seed, where the frozen arm's untouched kernel gives 0.0755 to 0.3480; and

one such partition, interleaving the nodes so that each group draws equally
 from all three clusters, leaves the ratio at 0.8071 to 3.0223, inside that family
 and on both sides of the no-preference line. The descent is opposed to the
 structure the environment carries rather than indifferent to structure. Two
 readouts do not carry the effect and we do not rest it on them. The initial-
 minus-final Frobenius distance to the template lies between -58.0962 and
 -48.1760 , so that distance grows; but the kernel norm grows by a factor of
 12.4094 to 14.6958 over the same interval, and a fixed total resource con-
 strains the sum of the entries rather than the norm, so most of that growth
 is concentration of mass onto fewer edges rather than loss of alignment. The
 edge-level correlation with the template ends between -0.0362 and -0.0208 ,
 near its starting value: at this sparsity the edge-level variance swamps a con-
 trast that the block means resolve. The template is a covariance of shared
 latent cluster signals, not an input to the update or a measured covariance
 of the constant frequencies.

The objective descends by most of what is available to it, coherence is
 maintained, and within-cluster coupling falls to a fraction of between-cluster
 coupling. Coherence is therefore not a proxy for representational alignment
 in this regime, and a plateau in the objective establishes neither stationar-
 ity of the adaptive process nor a thermodynamic entropy-production rate.
 Nor is the direction of the failure accidental. Symmetric coupling conserves
 the mean drift, so minimizing $\sum_i v_i^2$ can only equalize drift across nodes,
 and equalizing drift rewards coupling between clusters of unlike frequency—

407 the complement of a template that rewards coupling within clusters of like
 408 frequency. The objective and the template are in this pairing opposed by
 409 construction, and that is the point: nothing about a system’s dissipation
 410 fixes which structure its slow variables acquire. What generalizes beyond the
 411 pairing is that last sentence and not the sign of the effect, which is why the
 412 blind-partition reading matters—the descent locates the environment’s own
 413 structure and moves against it, rather than degrading any structure a reader
 414 might score it against. What fails here is the objective rather than ther-
 415 modynamically specified learning at large: an Ising model whose couplings
 416 are annealed degrees of freedom, driven by external fields that carry the
 417 data, memorizes and generalizes under ordinary statistical-mechanical rules
 418 (Corberi et al., 2026). Those couplings equilibrate under a Hamiltonian con-
 419 taining the data; these descend a squared-drift functional whose stationary
 420 point the data never enters. Which functional the slow variables follow is
 421 what decides whether they acquire environmental structure.

422 **4.4 The estimation protocol demonstrated on public** 423 **EEG**

424 Item 3 of Section 3.2 specifies estimators whose behavior on real recordings
 425 is not obvious, so we exercise them on public propofol-sedation EEG (Bajwa
 426 et al., 2025). The purpose is to show what the protocol produces and where
 427 it is fragile, not to test the awakening hypothesis, for which these data are

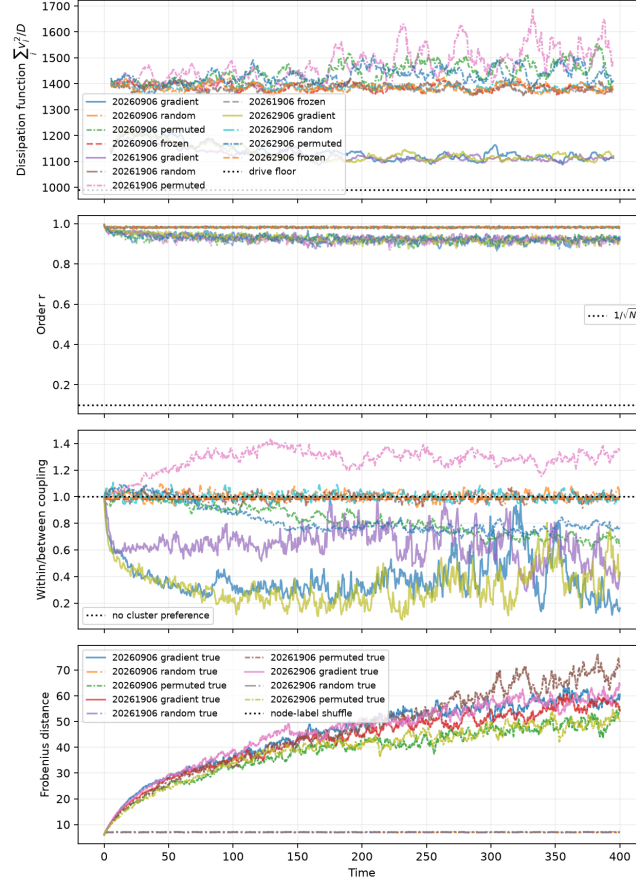


Figure 3: Joint fast/slow dynamics, four arms per seed on one phase-noise stream: gradient (solid), matched-norm random step (dash-dotted), the gradient step under a random node relabelling (dash-dot-dotted), frozen kernel (dashed). First: smoothed squared-drift dissipation against the drive floor the conserved mean drift imposes; only the gradient arm descends. Second: order and the finite-size reference. Third: mean within-cluster over mean between-cluster coupling, a measure unchanged by rescaling the kernel; the gradient arm falls well below the no-preference line, the relabelled arm stays above the gradient arm at the same kernel deformation, and the remaining arms stay on the line. Fourth: Frobenius distance to the true template (solid) and to a node-label shuffle (dotted), which track each other because the kernel norm, not the alignment, is what grows. Coherence persists while alignment is lost. The result limits the proposed connection between dissipation minimization and environmental structure.

428 the wrong design.

429 Two fragilities appear immediately. The implemented estimator pools 62
430 channels over a 100-ms window, so it estimates a spatiotemporal distribu-
431 tion rather than an instantaneous spatial one—a departure from the protocol
432 forced by channel count, and one that any scalp implementation will face.
433 Bipolar and common-reference montages give different residuals, making ref-
434 erencing a material analysis choice rather than a preprocessing detail.

435 The bipolar results are compatible with the Bessel relation over the ob-
436 served concentration range. That range is too narrow to distinguish the
437 curve from its linear approximation at the observed scatter, so the check is
438 uninformative about the relation itself. An informative test needs a broader
439 concentration range and explicit treatment of source mixing, finite samples
440 and temporal pooling; intracranial or high-density recordings across a gen-
441 uine state transition are the natural setting.

442 5 Thermodynamic constraints and the for- 443 mal composition

444 The synchronization model of Section 2 is one link of a longer conditional
445 construction, whose nine nodes are:

446 **n1** a finite phase space has bounded information capacity; **n2** a
447 broken symmetry forces a boundary, where the field leaves its vac-
448 uum manifold; **n3** the boundary erases, and erasure costs heat;

449 **n4** that dissipation bounds nonpredictive memory; **n5** discrete
450 couplings coarse-grain to a continuum limit; **n6** a specified phys-
451 ical field realizes that limiting coupling; **n7** above threshold the
452 field carries a unique coherent order parameter; **n8** compatible
453 local states glue to a unique global section; **n9** that section is the
454 unique fixed point of a self-prediction map.

455 Node-level results are established or cited; what the composition consumes
456 are the eight implications between them. Figure 4 shows where they sit and
457 what kind of assumption each one is, and Table 1 states them, classified by
458 their role in the composition rather than by their plausibility. In particular,
459 finite capacity does not imply a symmetry-breaking boundary: **n1** and **n2** are
460 independent roots, even though the composition can accept an implication
461 between them as an argument.

462 **5.1 Capacity, boundaries and predictive memory**

463 For a finite state space X , Shannon entropy satisfies $H(p) \leq \log |X|$, with
464 equality only for the uniform distribution. A stream with more possible
465 histories than $|X|$ cannot be encoded injectively in that register. Separately,
466 a continuous field on a connected substrate that reaches distinct components
467 of a disconnected vacuum manifold must leave that manifold somewhere,
468 the simplest topological-defect construction (Kibble, 1976). Relating this
469 locus to a particular finite information-processing register is the additional
470 hypothesis E12 carries.

471 A non-surjective update on a finite register is non-injective. Under the

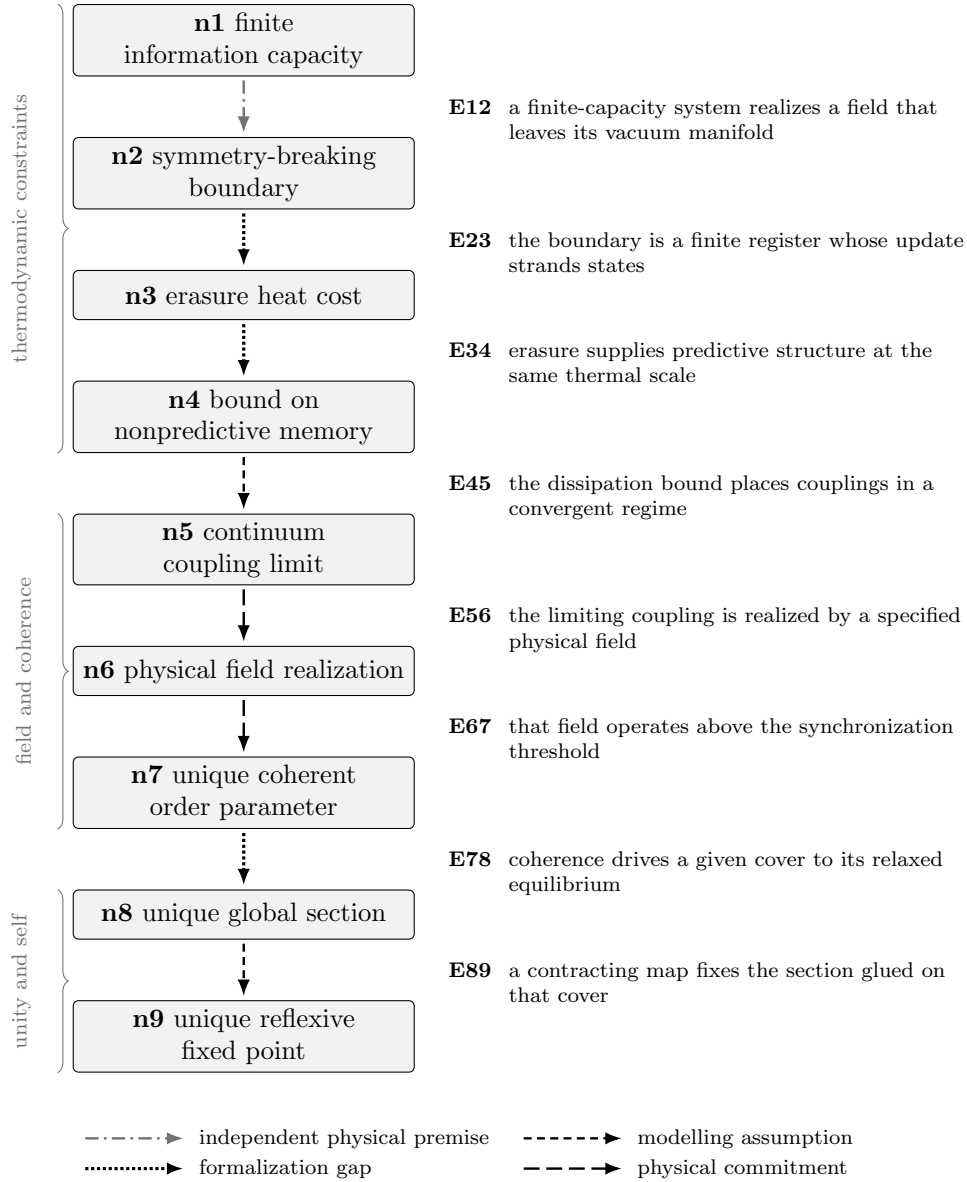


Figure 4: The conditional composition. Each node is an established or cited result; each arrow is an implication the composition consumes and no theorem here supplies, styled by the kind of assumption it carries. The three formalization gaps bracket the two physical commitments, so the chain’s empirical content sits between links of a different kind on either side. E12 is drawn as a premise because n1 and n2 are independent roots. Table 1 states each connection in full.

Edge	Status	Required connection
E12	Independent	A finite-capacity system realizes a field that leaves its vacuum manifold. Capacity alone does not imply this.
capacity $\rightarrow$ boundary	physical premise	
E23	Formalization	The boundary is associated with a finite register whose update leaves states unreachable.
boundary $\rightarrow$ erasure	gap	
E34	Formalization	Erasures supplies a predictive structure with the same thermal scale and heat budget, and nonzero wasted memory.
erasure $\rightarrow$ prediction	gap	
E45	Modelling	A predictive dissipation bound places coupling energies in a convergent coarse-graining regime.
prediction $\rightarrow$ continuum	assumption	
E56	Physical	The limiting coupling is realized by a specified field satisfying the kernel conditions. Its identification with cortex is empirical.
continuum $\rightarrow$ field	commitment	
E67	Physical	That field operates above the synchronization threshold.
field $\rightarrow$ coherence	commitment	
E78	Formalization	Coherence drives a cover, supplied as a separate argument of the theorem, to an equilibrium reached by relaxation at coupling at least the named mean-field value. That its patches carry compatible local states is a field of the cover rather than a conjunct of this edge.
coherence $\rightarrow$ unity	gap	
E89	Modelling	The state glued on that same cover is fixed by a contracting map mediated by its local self-encoding.
unity $\rightarrow$ self	assumption	

Table 1: The assumptions consumed by the conditional composition theorem, each an implication between two of the nine nodes above. Node-level results do not establish these connecting implications. The complete theorem map and witnesses are in Supplemental Material, Table S1.

472 specified system–bath model and thermal assumptions, erasure carries a Lan-
 473 dauer heat cost (Landauer, 1961). Finite capacity does not imply that every
 474 update erases, and logical many-to-one processing must be distinguished from
 475 its physical implementation (Norton, 2005; Shenker, 2000). The formaliza-
 476 tion records the physical heat identity as an obligation of each model and
 477 supplies an explicit one-bit eraser satisfying it.

478 For a system state X_t driven by an environmental signal S_t , the predictive
 479 dissipation bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (7)$$

480 Under the no-back-action Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$, data process-
 481 ing makes the bracket nonnegative (Still et al., 2012). The bound constrains
 482 nonpredictive memory at a given dissipation budget. It does not specify
 483 a learning rule, maximize prediction, or require the internal representation
 484 to reproduce environmental geometry—the point the plasticity control of
 485 Section 4 makes concretely. Connecting this budget to Landauer heat ad-
 486 ditionally requires nonpredictive information to be bounded by the entropy
 487 erased by the register. The formal bridge derives Eq. (7) from Landauer’s
 488 bound under that identification; it does not derive the identification.

489 A bound $D_{KL} \leq \Delta t \sigma$ also cannot force $D_{KL} \rightarrow 0$ unless its right-hand
 490 side tends to zero, and Gibbs’ inequality proves nonnegativity of relative en-
 491 tropy, not a general identification of relative entropy with physical entropy

production. The chain uses a different technical sense of information at each step—a counting bound on a register, a thermodynamic heat cost, a predictive mutual information—and those senses are not interchangeable (Rosas et al., 2026). These distinctions prevent an informational inequality from being read as a derivation of environmental learning, Bayesian inference, or semantic content.

5.2 Compatible states and reflexive fixed points

A sheaf formalizes local descriptions and their restriction to smaller regions. If local sections agree on every overlap, the sheaf condition gives a unique global section. In the probability-sheaf construction, this is consistency of local density information, and it is not implied by agreement of oscillator phases: the development includes a counterexample with equal phases but incompatible densities on an overlap. Section 6 rests on this distinction, and the model excludes such configurations only through its compatibility hypothesis.

A thermodynamic cover couples this gluing construction to a phase configuration at a potential minimum. For the deterministic positive-coupling model, convergence to that minimum is proved under an explicit initial basin condition; other equilibria prevent an unconditional conclusion. The stationary noisy order parameter, the deterministic relaxation, and the cover must therefore be connected by E78 rather than identified without a hypothesis. Neither field continuity nor synchronization alone guarantees a globally com-

514 patible representation.

515 A reflexive extension equips global sections with a complete metric and a
516 predictive map formed by restricting a state to a local encoding region and
517 reading that encoding out. If the map contracts and fixes the section supplied
518 by the cover, Banach’s theorem gives uniqueness of that same fixed point.
519 The contraction rate is assumed, and the equality between the glued section
520 and the fixed point is part of E89. A coherent order parameter supplies the
521 supercritical inequality required by the stated contraction bound; it does not
522 construct the map or prove that cortical dynamics implements it.

523 The theorem `chain` concludes `UnifiedSelf` from these hypotheses and
524 from two structures it also takes as arguments, a thermodynamic cover and
525 a reflexive boundary. The eight edges are not the whole of what is assumed:
526 the cover’s overlap-compatibility field—the load-bearing assumption of the
527 unity-is-not-synchrony argument—and the boundary’s resonance condition
528 are class fields, so they are consumed by the theorem without appearing
529 among its edges. A single witness satisfies all eight at once, so their conjunc-
530 tion is not empty; it is assembled from a double well, a register, a uniform-grid
531 mesh and a three-site cortex, which remain distinct toy systems rather than
532 one mechanism. The project declares no additional Lean axioms beyond
533 the foundational axioms used by Lean’s libraries; physical content resides
534 in hypotheses and model fields. This checks the conditional mathematical
535 composition, not the empirical truth of its premises. Detailed constructions,
536 scope statements and counterexamples are retained in the supplement and

537 source files.

538 6 What the construction commits to about 539 consciousness

540 The composition terminates in a mathematical object, and we take a posi-
541 tion on what that object is. The commitment has two clauses. *Unity*: the
542 unique global section glued from compatible local states is the formal corre-
543 late of the unity of a conscious episode—the fact that a subject undergoes
544 one experience rather than a collection of co-occurring local ones. *Selfhood*:
545 the fixed point of the self-prediction map on that same section is the formal
546 correlate of a minimal self, a state that is what it takes itself to be. Neither
547 clause is a theorem, and no simulation here supports either. We argue for
548 them, then say what would count against them.

549 Three features of the construction motivate the identification, and each
550 distinguishes it from a bare synchrony measure.

551 First, compatibility rather than agreement. Oscillator phases can agree
552 everywhere while the local probabilistic descriptions they carry remain in-
553 compatible on their overlaps, and Section 5.2 exhibits that configuration
554 inside the formalization, not only as an illustration. If the unity of experi-
555 ence is a matter of local contents being about one world rather than merely
556 co-timed, then compatibility, not coherence, is the property to look for—and
557 coherence enters as the physical condition under which compatibility can be

558 maintained, which is what E78 asserts.

559 Second, uniqueness follows from the construction rather than from a sep-
560 arate selection principle. Accounts of conscious unity need something that
561 picks one experience from among the candidates a system could support;
562 integrated information theory postulates exclusion for this purpose (Tononi
563 et al., 2016). Here uniqueness arrives twice as a consequence: given a cover
564 whose local sections agree, the sheaf condition gives exactly one global sec-
565 tion, and given a contracting self-prediction map, Banach’s theorem gives
566 exactly one fixed point. The scope of that claim needs stating exactly, be-
567 cause the stronger reading is available and is not ours. Uniqueness is relative
568 to a cover and a map, and choosing them—which regions, which overlaps,
569 which encoding region, and so where the subject ends—is not supplied by ei-
570 ther theorem. It is what E78 and E89 carry. What the construction removes
571 is the need for an additional principle to select among the experiences a fixed
572 cover and map would otherwise admit; it does not remove the need to say
573 which cover and map a given subject has. That is a structural argument,
574 not evidence.

575 Third, the reflexive clause makes the self a condition rather than an addi-
576 tion. The predictive map restricts a global state to a region encoding it and
577 reads that encoding back out, so a fixed point is a state whose self-model is
578 accurate about the state it models. That condition is available only because
579 the state modelled is a single glued object: a system whose local descrip-
580 tions do not cohere has no such state for a self-model to be about. Unity

581 and selfhood are therefore one construction read at two levels, which is why
582 `UnifiedSelf` requires the fixed point to be the glued section itself rather
583 than an unrelated section.

584 The costs belong with the argument. The commitment does not address
585 why any physical condition is accompanied by experience; it is a proposal
586 about which physical state is the conscious one, not about why states of that
587 kind feel like anything. It is therefore open to the objection that a system
588 could satisfy every condition—compatible local states, a supercritical field,
589 an accurate self-model—while nothing is experienced. We have no reply to
590 that objection and claim none. The identification is also multiply realizable
591 as stated: nothing in the sheaf or fixed-point conditions is specific to cortex
592 or to electromagnetic coupling, so the substrate claim is carried entirely by
593 E56 and E67 and is empirical. No conclusion about artificial systems follows
594 from the formal results.

595 What could count against the commitment is dissociation. It predicts that
596 the two clauses and the presence of a unified conscious episode go together,
597 so a state in which local cortical descriptions are demonstrably compatible
598 and the field supercritical, while the subject reports no unified experience,
599 would tell against it; so would unified reported experience where compati-
600 bility demonstrably fails. Anesthetic emergence, dreaming and dissociative
601 states are the places to look, because they separate coherence from report
602 in ways ordinary waking does not. Such tests need an operational measure
603 of overlap compatibility on neural data, which this article does not supply;

604 constructing one is the main obstacle to making the commitment empirically
605 live rather than merely stated. Supplemental Material specifies what such an
606 observable would have to satisfy—a restriction map, a statistic with a null
607 under decoding noise, validation against configurations whose compatibility
608 is built rather than inferred—and the three ways it would fail silently, one
609 of which is reading compatibility off synchrony and so assuming away the
610 distinction being tested.

611 The physical tests of Sections 3 and 4 are independent of all of this. They
612 concern coupling, coherence and recovery, and can fail while the identification
613 stands, or succeed while it is wrong.

614 **7 Discussion**

615 The framework connects thermodynamic information constraints, collective
616 phase organization and compatible local representations, conditionally and
617 with its assumptions named. Its biological content is a coupling-gated re-
618 covery hypothesis with defined observables, a stated protocol, and failure
619 conditions declared in advance. Its formal content is a composition whose
620 eight gaps are exhibited rather than argued past, and which can be inhabited
621 by a single model. Its interpretive content is the identification of Section 6,
622 offered as a commitment with reasons.

623 The simulations are useful because they restrict the hypothesis rather
624 than support it. Finite ramps smear the stationary onset to the point where

625 a fitted onset exponent chiefly reports the ramp; an unpinned collective
626 phase makes independence diagnostics ambiguous in exactly the supercritical
627 regime the model cares about; and squared-drift plasticity maintains coher-
628 ence while degrading structural alignment. The second and third of these
629 are results about method and about a common intuition, and they transfer
630 beyond this framework.

631 The proposal differs in emphasis from accounts based on integrated in-
632 formation (Tononi et al., 2016), global workspace dynamics (Baars, 1997;
633 Dehaene et al., 2014), or predictive processing (Friston, 2010). It specifies a
634 candidate interaction medium and a mathematical compatibility condition,
635 and, relative to a given cover and self-prediction map, it derives the unique-
636 ness those accounts postulate. It does not derive the exclusion principle of
637 integrated information, identify every broadcast with a field transition, or
638 derive a generative model from a dissipation bound. Continuous fields can
639 carry incompatible local descriptions and discrete models can admit sheaf
640 constructions, so continuity alone does not settle the combination problem;
641 compatibility is what this account offers in its place.

642 Two formal results bear on the substrate question, and both admit a
643 stronger reading than they support. At matched coupling resource, a fixed
644 wiring support that excludes a better-correlated pair can be suboptimal—a
645 statement about constrained optimization, not about biology. On an atom-
646 less substrate, a kernel supported on finitely many points contributes zero to
647 the chosen continuum coupling integral. That is a property of the measure

648 and the observable, and it cuts against this framework’s own coarse-graining
649 step as much as against anything else. It does not imply that digital hard-
650 ware cannot approximate a field or support consciousness, and no biological-
651 versus-artificial conclusion follows.

652 The empirical priorities follow directly from Section 3.2: estimate effec-
653 tive coupling and diffusion in consistent units, test the proposed relation
654 between coupling and extracellular geometry, and measure recovery against
655 an independently defined behavioral endpoint under matched model compar-
656 ison. A favorable concentration–coherence relation would support a statis-
657 tical description, not identify its physical cause; an unfavorable result in a
658 regime satisfying the stated assumptions would constrain that description.
659 Successful synchronization simulations cannot substitute for measurements
660 distinguishing ephaptic interactions from other coupling mechanisms.

661 The article therefore asks to be judged in three separable parts: a pre-
662 diction that can be tested, a composition that can be checked, and an iden-
663 tification that can be argued about. Keeping them separate is what makes
664 the whole assessable.

665 **Data and code availability**

666 The Lean development, simulation scripts, tests and saved numerical sum-
667 maries are maintained in the manuscript source repository. Publication sim-
668 ulation values are regenerated from saved summaries without rerunning pro-

duction sweeps. The exploratory EEG analysis uses OpenNeuro dataset
ds005620 (Bajwa et al., 2025); preprocessing and selection are described
in Supplemental Material. The source repository contains the reproducible
analysis and does not redistribute the raw recordings.

Funding and competing interests

This research received no external funding. The author declares no competing interests.

Use of AI tools

AI assistance to this work was substantial and is not separable by task. Several coding assistants and chat interfaces were used in an interleaved way throughout: Google’s Antigravity CLI (Gemini 3.1, Claude 4.6 Opus, Claude 4.6 Sonnet), OpenAI’s Codex CLI (Astra, Terra, Luna, Sol), Claude Code (Claude Opus 5), Gemini Pro through its web interface, and Fable (Claude Opus 4) through OpenRouter, accessed with the Hermes CLI. Between them they contributed to the Lean development, the simulation and analysis code, and the drafting and revision of both documents. Because the tools were used in combination and in alternation, no section, proof, figure or result is attributable to any one of them, and none is claimed to be. The author is responsible for all content, including everything a tool produced. Machine

688 checking establishes the stated Lean results under their hypotheses; it does
689 not validate biological interpretations or replace scientific review of generated
690 code and prose.

691 References

692 Pinotsis, D. A., & Miller, E. K. (2022). Beyond dimension reduction: Stable
693 electric fields emerge from and allow representational drift. *NeuroImage*,
694 253, 119058.

695 Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated informa-
696 tion theory: from consciousness to its physical substrate. *Nature Reviews*
697 *Neuroscience*, 17(7), 450-461.

698 Baars, B. J. (1997). In the Theater of Consciousness: The Workspace of the
699 Mind. *Oxford University Press*.

700 Dehaene, S., Charles, L., King, J. R., & Marti, S. (2014). Toward a compu-
701 tational theory of conscious processing. *Current Opinion in Neurobiology*,
702 25, 76-84.

703 Friston, K. (2010). The free-energy principle: a unified brain theory?. *Nature*
704 *Reviews Neuroscience*, 11(2), 127-138.

705 Pinotsis, D. A., & Miller, E. K. (2023). In vivo ephaptic coupling allows
706 memory network formation. *Cerebral Cortex*, 33(17), 9877-9890.

707 Pinotsis, D. A., Fridman, G., & Miller, E. K. (2023). Cytoelectric coupling:
 708 Electric fields sculpt neural activity and "tune" the brain's infrastructure.
 709 *Progress in Neurobiology*, 226, 102465.

710 McFadden, J. (2002). The conscious electromagnetic information (cemi) field
 711 theory: the hard problem made easy?. *Journal of Consciousness Studies*,
 712 9(8), 45-60.

713 Pockett, S. (2012). The electromagnetic field theory of consciousness: a
 714 testable hypothesis about the characteristics of conscious as opposed to
 715 non-conscious fields. *Journal of Consciousness Studies*, 19(11-12), 191-223.

716 McFadden, J. (2020). Integrating information in the brain's EM field: the
 717 cemi field theory of consciousness. *Neuroscience of Consciousness*, 2020(1),
 718 niaa016.

719 Fröhlich, F., & McCormick, D. A. (2010). Endogenous electric fields may
 720 guide neocortical network activity. *Neuron*, 67(1), 129-143.

721 Anastassiou, C. A., & Koch, C. (2015). Ephaptic coupling to endogenous
 722 electric field activity: why bother?. *Current Opinion in Neurobiology*, 31,
 723 95-103.

724 Pockett, S. (2002). Difficulties with the electromagnetic field theory of con-
 725 sciousness. *Journal of Consciousness Studies*, 9(4), 51-56.

726 Vöröslakos, M., et al. (2018). Direct effects of transcranial electric stimulation
 727 on brain circuits in rats and humans. *Nature Communications*, 9, 483.

728 Verkhatsky, A., & Nedergaard, M. (2018). Physiology of Astroglia. *Physio-*
729 *logical Reviews*, 98(1), 239-389.

730 Bastos, A. M., Donoghue, J. A., Brincat, S. L., Mahnke, M., Yanar, J., Cor-
731 rea, J., ... & Miller, E. K. (2021). Neural effects of propofol-induced uncon-
732 sciousness and its reversal using thalamic stimulation. *eLife*, 10, e60824.

733 Bajwa, I. J., Nilsen, A. S., Skukies, R., Aamodt, A., Ernst, G., Storm, J.
734 F., & Juel, B. E. (2025). A repeated awakening study exploring the capac-
735 ity of complexity measures to capture dreaming during propofol sedation.
736 *Scientific Reports*, 15, 32012.

737 Xie, L., Kang, H., Xu, Q., Chen, M. J., Liao, Y., Thiyagarajan, M., ... &
738 Nedergaard, M. (2013). Sleep drives metabolite clearance from the adult
739 brain. *Science*, 342(6156), 373-377.

740 Syková, E., & Nicholson, C. (2008). Diffusion in brain extracellular space.
741 *Physiological Reviews*, 88(4), 1277-1340.

742 Barbour, B. (2017). Analysis of claims that the brain extracellular impedance
743 is high and non-resistive. *Biophysical Journal*, 113(7), 1636-1638.

744 Norton, J. D. (2005). Eaters of the lotus: Landauer's principle and the return
745 of Maxwell's demon. *Studies in History and Philosophy of Science Part B:*
746 *Studies in History and Philosophy of Modern Physics*, 36(2), 375-411.

747 Landauer, R. (1961). Irreversibility and heat generation in the computing
748 process. *IBM Journal of Research and Development*, 5(3), 183-191.

749 Kibble, T. W. B. (1976). Topology of cosmic domains and strings. *Journal*
750 *of Physics A: Mathematical and General*, 9(8), 1387-1398.

751 Sakaguchi, H. (1988). Cooperative phenomena in coupled oscillator systems
752 under external fields. *Progress of Theoretical Physics*, 79(1), 39-46.

753 Still, S., Sivak, D. A., Bell, A. J., & Crooks, G. E. (2012). Thermodynamics
754 of prediction. *Physical Review Letters*, 109(12), 120604.

755 Kawai, R., Parrondo, J. M. R., & Van den Broeck, C. (2007). Dissipation:
756 The phase-space perspective. *Physical Review Letters*, 98(8), 080602.

757 Corberi, F., dello Russo, S., Messuti, G., Scarpetta, S., & Smaldone, L.
758 (2026). Thermodynamic learning. *arXiv preprint*. arXiv:2609.04732.

759 Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur ap-
760 plication aux équations intégrales. *Fundamenta Mathematicae*, 3(1), 133-
761 181.

762 Shenker, O. R. (2000). Logic and entropy. *PhilSci-Archive preprint*. philsci-
763 archive.pitt.edu/115.

764 Rosas, F. E., Mediano, P. A. M., Luppi, A. I., Boyd, A. B., Berta, M., Jensen,
765 H. J., Gastpar, M., & Carroll, S. M. (2026). Four faces of information in
766 natural systems. *Nature Reviews Physics*. doi:10.1038/s42254-026-00971-4.

767 Sznitman, A.-S. (1991). Topics in propagation of chaos. In *École d'Été de*

- 768 *Probabilités de Saint-Flour XIX—1989*, Lecture Notes in Mathematics,
769 vol. 1464, pp. 165–251. Springer. doi:10.1007/BFb0085169.
- 770 Townsend, R. G., Solomon, S. S., Chen, S. C., Pietersen, A. N. J., Martin,
771 P. R., Solomon, S. G., & Gong, P. (2015). Emergence of complex wave
772 patterns in primate cerebral cortex. *The Journal of Neuroscience*, 35(11),
773 4657-4662. doi:10.1523/JNEUROSCI.4509-14.2015.
- 774 Xu, Y., Long, X., Feng, J., & Gong, P. (2023). Interacting spiral wave pat-
775 terns underlie complex brain dynamics and are related to cognitive pro-
776 cessing. *Nature Human Behaviour*, 7(7), 1196-1215. doi:10.1038/s41562-
777 023-01626-5.